

Multi-Cloud Disaster Recovery System

System Architecture Document

Multi-Region, Multi-Cloud DR Architecture

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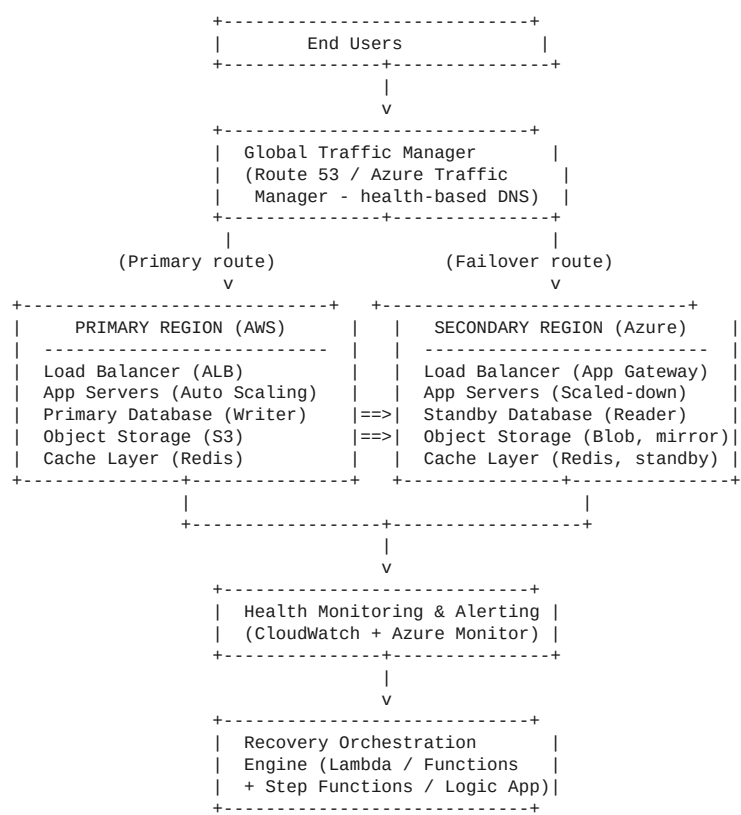
Industry Skills: AWS - Azure - Disaster Recovery - Automation - High Availability

1. Architecture Overview

The system follows a warm-standby, multi-region disaster recovery architecture. A primary (active) region on AWS serves live production traffic, while a secondary (standby) region on Azure (or a second AWS region) runs a scaled-down but fully deployed replica of the application stack. Data is continuously replicated from primary to secondary. A dedicated monitoring and orchestration layer watches the health of both regions and automatically redirects traffic and promotes the standby environment when the primary becomes unhealthy.

2. High-Level Architecture Diagram (Textual)

The diagram below represents the logical flow of traffic, replication, and control signals across the two regions.



3. Core Architectural Components

Component	Purpose	Example Services
Global Traffic Manager	DNS/traffic-level routing with health checks; redirects users to healthy region	AWS Route 53 (health checks + failover routing), Azure Traffic Manager
Compute / App Tier	Runs application workloads; standby scales up on failover	AWS EC2/ECS/EKS Auto Scaling, Azure VM Scale Sets/AKS

Component	Purpose	Example Services
Database Tier	Primary-replica database replication across regions	Amazon Aurora Global Database / RDS Cross-Region Read Replica, Azure SQL Geo-Replication, or PostgreSQL logical replication
Object / Blob Storage	Stores files, backups, static assets; cross-region replicated	S3 Cross-Region Replication, Azure Blob geo-redundant storage (GRS)
Cache Layer	Reduces DB load; rebuilt/replicated in standby region	Amazon ElastiCache (Redis), Azure Cache for Redis
Health Monitoring	Collects metrics/logs; evaluates region health signals	Amazon CloudWatch, Azure Monitor, synthetic uptime checks
Recovery Orchestration Engine	Executes failover/failback workflow logic	AWS Step Functions + Lambda, Azure Logic Apps + Functions
Backup & Snapshot Service	Scheduled backups with retention and restore testing	AWS Backup, Azure Backup, database-native snapshotting
Infrastructure as Code	Ensures identical, repeatable infra in both regions	Terraform / AWS CloudFormation / Azure Bicep
Notification Layer	Alerts stakeholders on health events and failover actions	Amazon SNS, Azure Notification Hubs, Slack/Email/PagerDuty webhooks

4. Data Replication Architecture

Database replication uses an asynchronous, near-real-time approach to minimize primary-region write latency while keeping RPO under one minute. Two supported patterns:

- **Managed cross-region replication** - e.g., Aurora Global Database or Azure SQL auto-failover groups, which handle log shipping and promotion natively.
- **Custom logical replication** - e.g., PostgreSQL logical replication or Debezium-based CDC (Change Data Capture) streaming into the secondary region's database, used when a fully cloud-agnostic setup is required.

Object storage uses native cross-region/cross-cloud replication (S3 CRR to a bucket mirrored into Azure Blob via a scheduled sync job, or direct GRS on Azure). Replication lag is exported as a metric and monitored continuously; alerts fire if lag exceeds the RPO threshold.

5. Failover Orchestration Flow

- **Step 1 - Detect:** Health monitors run synthetic checks and infrastructure-level probes every 10-15 seconds across compute, database, and network layers in the primary region.
- **Step 2 - Confirm:** A failure must be confirmed by multiple consecutive failed checks (e.g., 3 checks over 45 seconds) to avoid false positives, following a quorum/consensus rule.
- **Step 3 - Decide:** The orchestration engine evaluates confirmed health state and decides whether to trigger automated failover, or escalate for manual approval based on configured policy.

- **Step 4 - Promote:** The standby database is promoted to primary (writer) role; replication direction is paused/reversed.
- **Step 5 - Scale:** Standby compute auto-scales to production capacity (pre-warmed or scale-out from minimal baseline).
- **Step 6 - Redirect:** DNS/traffic manager updates routing so new requests resolve to the secondary region.
- **Step 7 - Notify:** Stakeholders receive real-time alerts with incident details and dashboard links.
- **Step 8 - Failback:** Once the primary region is verified healthy, data is re-synced and traffic is gradually shifted back following a controlled failback runbook.

6. Network & Security Architecture

- Private connectivity between regions/clouds via VPN or dedicated interconnect (e.g., AWS Site-to-Site VPN / Azure VPN Gateway, or ExpressRoute/Direct Connect for higher-throughput replication).
- Isolated VPC/VNet per region with least-privilege security groups/NSGs.
- Centralized IAM policy design mirrored across AWS IAM and Azure RBAC for consistent access control.
- All data encrypted in transit (TLS 1.2+) and at rest (KMS / Azure Key Vault-managed keys).
- Secrets management via AWS Secrets Manager / Azure Key Vault, replicated to both regions.

7. Deployment Model

Model	Description	Used For
Active-Passive (Warm Standby)	Primary serves all traffic; secondary runs at reduced capacity and scales up on failover.	Default/baseline model - cost-efficient
Active-Active (Optional / Advanced)	Both regions serve live traffic concurrently behind global load balancing.	Highest availability tier for critical workloads
Pilot Light	Only core data services replicated; compute provisioned on-demand during failover.	Lower-priority or cost-sensitive workloads